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Programme Master

Programme Code MCAOL

Course Tittle Cloud and Data Science Lab

Course Code MCS-229

Regional Centre DEHRADUN

Study Centre Online

Assignment No. MCA_NEW(III)/229/Assign/2025

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Date: 20 / 10 / 2025

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Course Code	:	MCSL-229
Course Title	:	Cloud and Data Science Lab
Assignment Number	:	MCA_NEW(III)/L-229/Assign/2025
Maximum Marks	:	100
Weightage	:	30%
Last Dates for Submission	:	30 th April, 2025 (for January session) 31 st October, 2025 (for July session)

The assignment has two sections. Answer all the questions. Each section is for 20 marks. The lab record of the Cloud Computing Lab and Data Science lab carries 20 Marks each. The remaining 20 marks are for viva voce. You may use illustrations, diagrams and screenshots to enhance the explanation. Please go through the guidelines regarding assignments given in the MCA(New) Programme Guide for the format of the presentation. If any assumptions are made, please state them.

SECTION-I: Cloud Computing Lab

Q1: (4+3+3 = 10 Marks)

- (a) Use Google Docs to create a document containing the schedule of MCS-229 practical counselling sessions. Store it on Google Drive and share it with five friends who can view and Comment on it.
- (b) Use Google Sheets and create a spreadsheet containing the yearly Salary details of 10 employees of a university under the headings – employee name, yearly basic salary, DA percentage, Provident fund deductions, Income Tax deduction, and net salary. You may use the following formulas for making the spreadsheet:
 - DA amount = DA percentage * yearly basic salary
 - Provident fund = 8% of yearly basic salary
 - Income Tax deduction is computed as:
 - For yearly basic salary < 5,00,000 tax =0
 - For yearly basic salary >= 5,00,000 but <10,00,000
tax = 20% of (yearly basic salary – 5,00,000)
 - For yearly basic salary >= 10,00,000
tax = 1,00,000 + 30% of (yearly basic salary – 10,00,000)
 - Net Salary = yearly basic salary + DA amount – Provident fund – tax
- (c) Use Google Slides and prepare nine slides on the topic “Platform as a Service (PaaS)” in a group of three students by sharing the Google Slides in your group in *edit* mode. Every group member should make three slides each and contribute to the slides of other members of her/his group.

Q2: (5 Marks)

Explore JustCloud file storage solutions and cloud storage on AWS. Use only the trial versions.

Q3: (5 Marks)

Use Google App Engine to write a program to multiply two matrices. Deploy it on the Google cloud.

SECTION-II: Data Science Lab

Q1:

(2+2+4+2 = 10 Marks)

The height of 50 students of class X was measured in centimeters. The following table shows this data. Perform the tasks (i) to (iv) using R programming.

156	155	163	151	141	160	151	131	170	163
165	160	158	168	158	154	162	166	171	141
161	159	158	156	149	145	168	171	165	145
166	167	169	156	164	146	155	164	168	172
156	163	167	157	155	145	171	164	166	195

- (i) Find the minimum and maximum height.
- (ii) Find the percentage of students whose height is between 160 and 170 Centimeters.
- (iii) Create and draw the frequency distribution with the help of a relevant graph.
- (iv) Find the outlier of the data.

Q2:

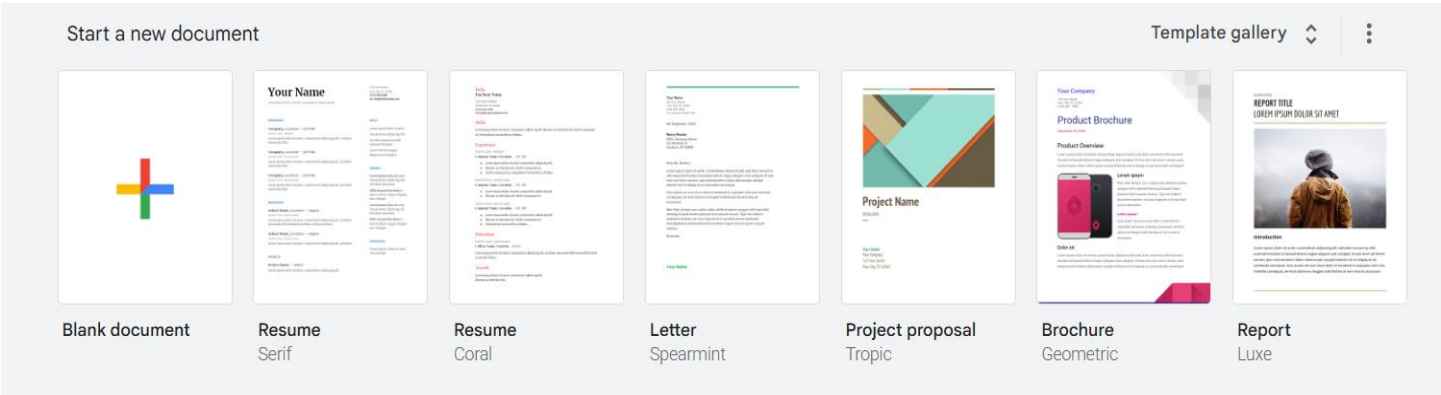
(10 Marks)

An organisation collected the following data of its employees: “years in employment” and “employee's salary”. Use R programming to fit a linear regression line to predict the effect of years in employment on the employee's salary. Also, predict the salary of an employee who is employed for 11 years.

Employee Number	Years in Employment	Employee Monthly salary in thousand Rs
1	10	149
2	8	120
3	12	150
4	7	110
5	6	100
6	5	90
7	9	130
8	8	110
9	3	80
10	4	85

Q1: (a) : Use Google Docs to create a document containing the schedule of MCS-229 practical counselling sessions. Store it on Google Drive and share it with five friends who can view and Comment on it.

Step1. Go to docs.google.com and click on “blank document”.

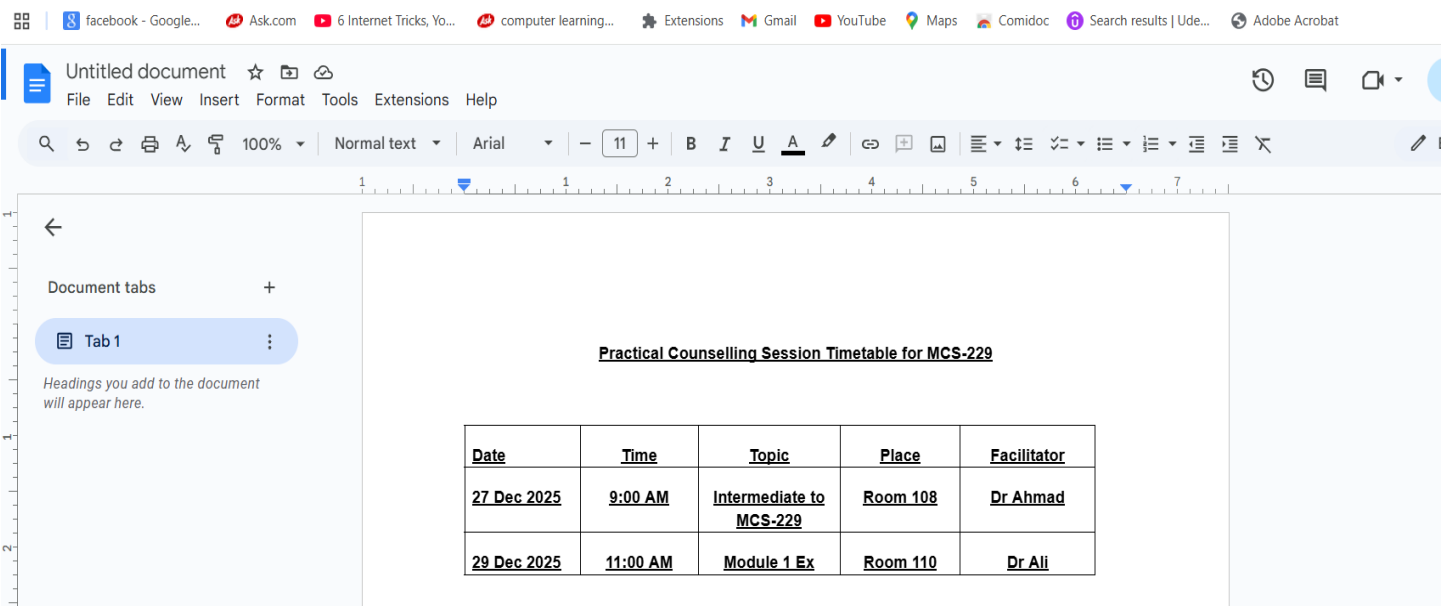


Step2. : Create the document titled “Schedule of MCSL-229 Practical Counselling Session “.

Practical Counselling Session Timetable for MCS-229

<u>Date</u>	<u>Time</u>	<u>Topic</u>	<u>Place</u>	<u>Facilitator</u>
<u>27 Dec 2025</u>	<u>9:00 AM</u>	<u>Intermediate to MCS-229</u>	<u>Room 108</u>	<u>Dr Ahmad</u>
<u>29 Dec 2025</u>	<u>11:00 AM</u>	<u>Module 1 Ex</u>	<u>Room 110</u>	<u>Dr Ali</u>

Step3: click on ‘File -> Download to save a copy.



Step4: Open Google Drive and upload the saved document.

The screenshot shows the Google Drive web interface. On the left is a sidebar with navigation options: Home, My Drive, Computers, Shared with me, Recent, and Spam. The main area displays 'Welcome to Drive' with 'Suggested folders' (MCSA-Afghan-Telecom, RHCA-Linux, Linux, IBR System Tutorials) and 'Suggested files' (Untitled document). Below this is the 'Recent' section with filters for Type, People, Modified, and Source. The 'Recent' list shows a file named 'MCS-229.docx' owned by 'me' and located in 'My Drive'.

Step5: Right- click on the file in Drive and choose “Share”.

Step6: click on send to share the file with your friends.

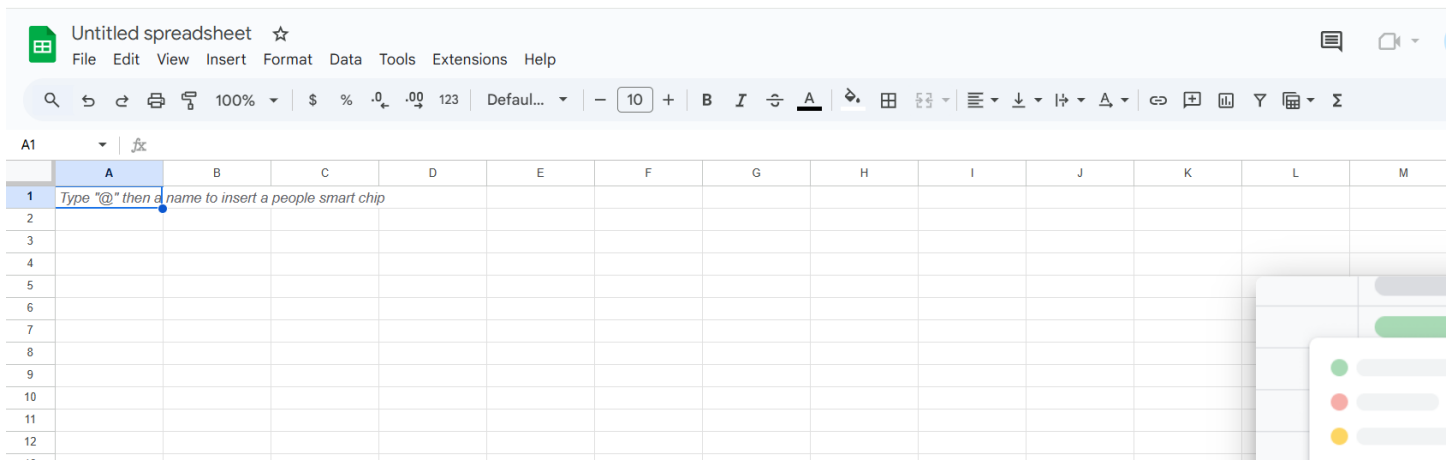
(b) : Use Google Sheets and create a spreadsheet containing the yearly Salary details of 10 employees of a university under the headings – employee name, yearly basic salary, DA percentage, Provident fund deductions, Income Tax deduction, and net salary. You may use the following formulas for making the spreadsheet:

DA amount = DA percentage * yearly basic salary
 Provident fund = 8% of yearly basic salary
 Income Tax deduction is computed as:
 For yearly basic salary < 5,00,000 tax = 0
 For yearly basic salary >= 5,00,000 but <10,00,000 tax = 20% of (yearly basic salary – 5,00,000)
 For yearly basic salary >= 10,00,000 tax = 1,00,000 + 30% of (yearly basic salary – 10,00,000)
 Net Salary = yearly basic salary + DA amount – Provident fund – tax

Step1: Go to docs.google.com the select “sheets”.

The screenshot shows the Google Sheets web interface. At the top, there's a 'Start a new spreadsheet' section with a 'Blank spreadsheet' option. To the right is a 'Template gallery' with various templates: To-do list, Annual budget, Monthly budget, Google Finance Invest..., and Annual Calendar. At the bottom, there's a navigation bar with 'Earlier', 'Owned by anyone', 'Last opened by me', and a date 'Aug 9, 2022'.

Step2 : Click on “ Blank Spreadsheet”.



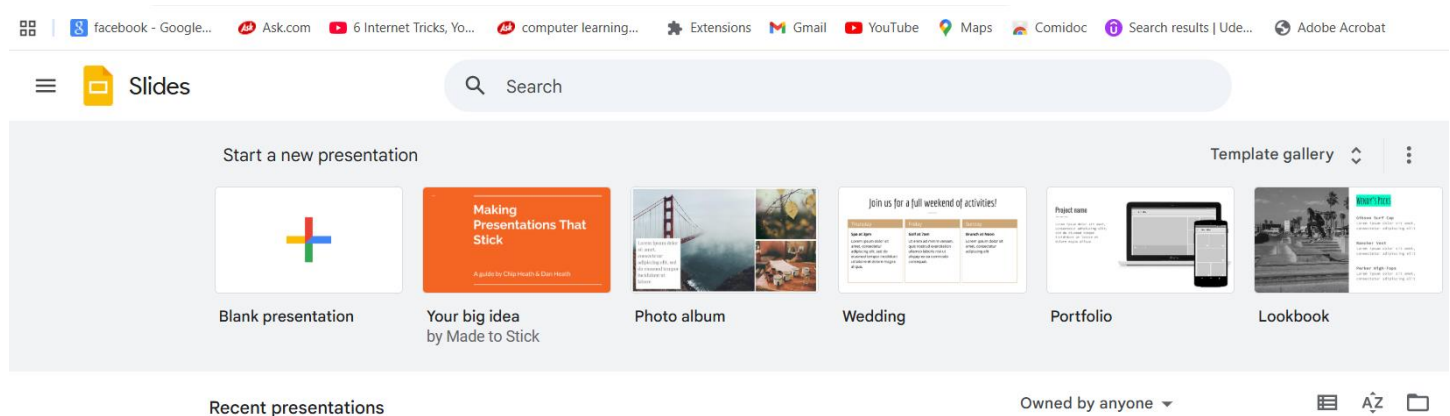
Step3: Create a spreadsheet with 10 employees including columns for Employee Name, Yearly Basic salary, DA% DA amount , provide fund , income tax, and Net salary, Use formulas:

- DA Amount <\$500000 -> Tax = 0
- If \$500000 <= Salary < \$1000000 -> Tax = 20% of (Salary - \$500000)
- If salary >= \$1000000 -> Tax = \$ 100000 + 30% of (salary - \$1000000)
- Net salary = Yearly Basic + DA – PF – Tax

	A	B	C	D	E	F	G	H	I	J
1	Employee Name	Yearly Basic Sal	DA percentage	DA Amount	Provided Fund	Income Tax Deduction	Net salary			
2	Jan	320000	11%	35200	25000	5000	332200			
3	Khan	460000	13%	59800	37000	0	521800			
4	ALi	620000	14%	86800	50000	25000	661800			
5	Ahmad	820000	9%	73800	66000	55000	777800			
6	Noor	990000	18%	178200	85000	95000	1074200			
7	Jan ali	210000	11%	23100	18000	0	231000			
8	khan ali	570000	13%	74100	46000	12000	618100			
9	wasel	1250000	26%	325000	99000	165000	1400000			
10	momin	730000	16%	116800	58000	47000	789800			
11	Nab	410000	11%	45100	34000	0	421100			
12										
13										

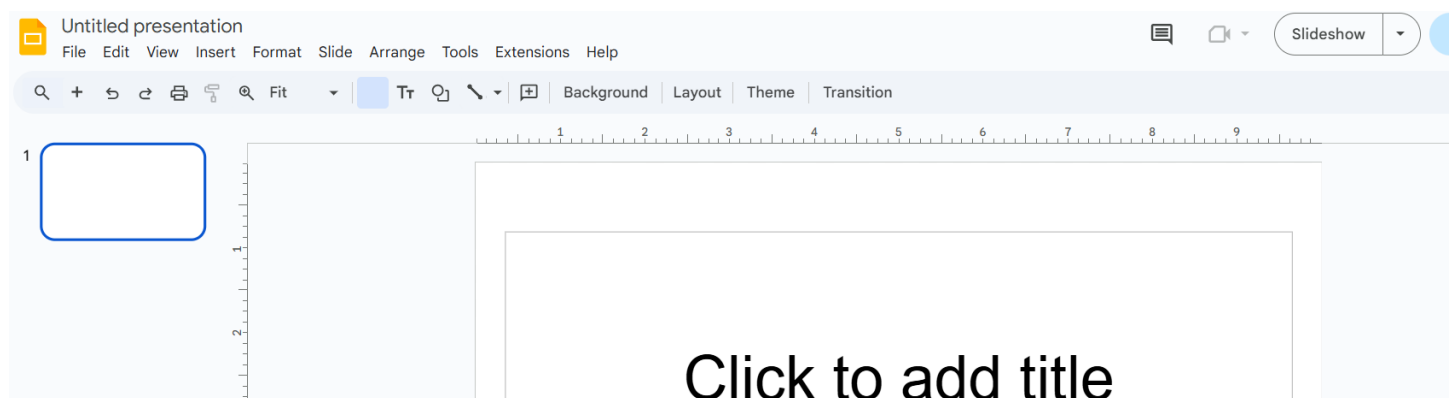
(c) : . Use Google Slides and prepare nine slides on the topic “Platform as a Service (PaaS)” in a group of three students by sharing the Google Slides in your group in edit mode. Every group member should make three slides each and contribute to the slides of other members of her/his group.

Step1: open www.docs.google.com



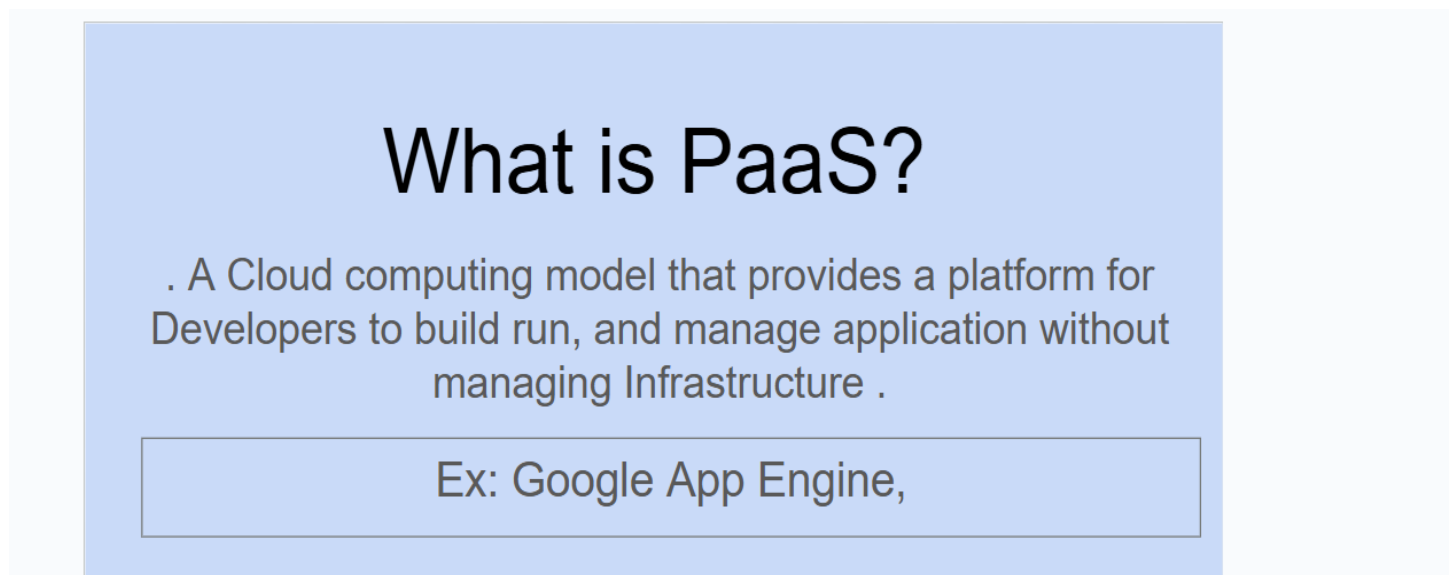
Step2: Open Slides:

Step 3: Click on blank presentation.



Step4: Now select them and create slides of “Platform as a Service (PaaS)”. (9 Slide).

Slide1:



Slide2:

Key Features of PaaS

- > Development Tools
- > Middleware
- > Database Managment
- > Application Hosting

Slide3:

Advantages of PaaS

- <> Faster Development
- <> Reduced Coding Effort
- <> Multi-plateform
- <> Cost- Effective

Slide4:

Disadvantage Of PaaS

- <> Vendor Lock in
- <> Limited infrastructure Control
- <> Security and Compliance Concerns

Slide5:

Popular PaaS Providers

- <> Google app engine
- <> Ms Azure
- <> AWS Elastic beanstalk
- <> Heroku

Slide6:

Use Cases of PaaS

- <> web Application development
- <> API management.
- <> Real time data Apps.

Slide7:

PaaS Vs LaaS vs SaaS

- <> Comparison Table.
- <> LaaS: Full control over infrastructure.
- <>: PaaS : Focus on development
- <> SaaS: Ready - to - Use software.

Slide8:

Security in PaaS

1. Common security risks.
2. Measures to ensure security.

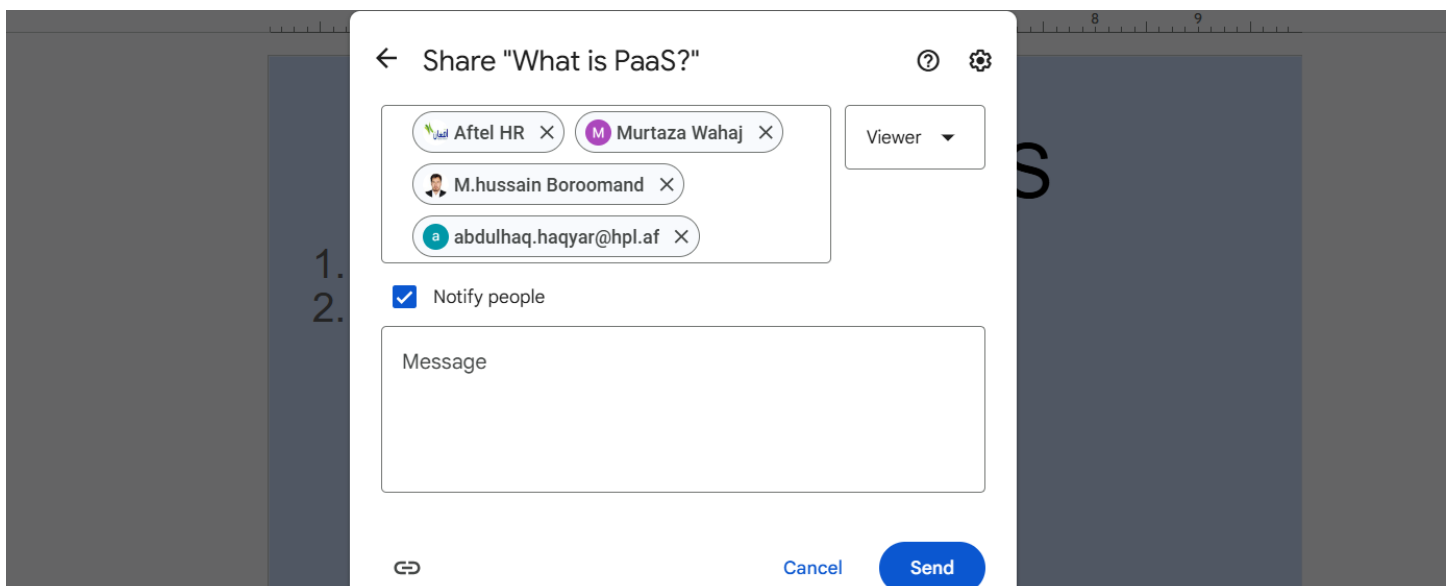
Slide9:

Conclusion

- <> PaaS simplifies application development
- <> Great for teams focusing on app logic over infrastructure.

Thank You!!!!!!

Step5: Now Share this presentation of three students in edit mode. Each student can create their slides based on their assigned topics and then the group can collaborate by reviewing and editing each other's.



Q2: Explore JustCloud file storage solutions and cloud storage on AWS. Use only the trial versions.

Ans:

Exploring Justcloud and AWS cloud storage storage solutions using

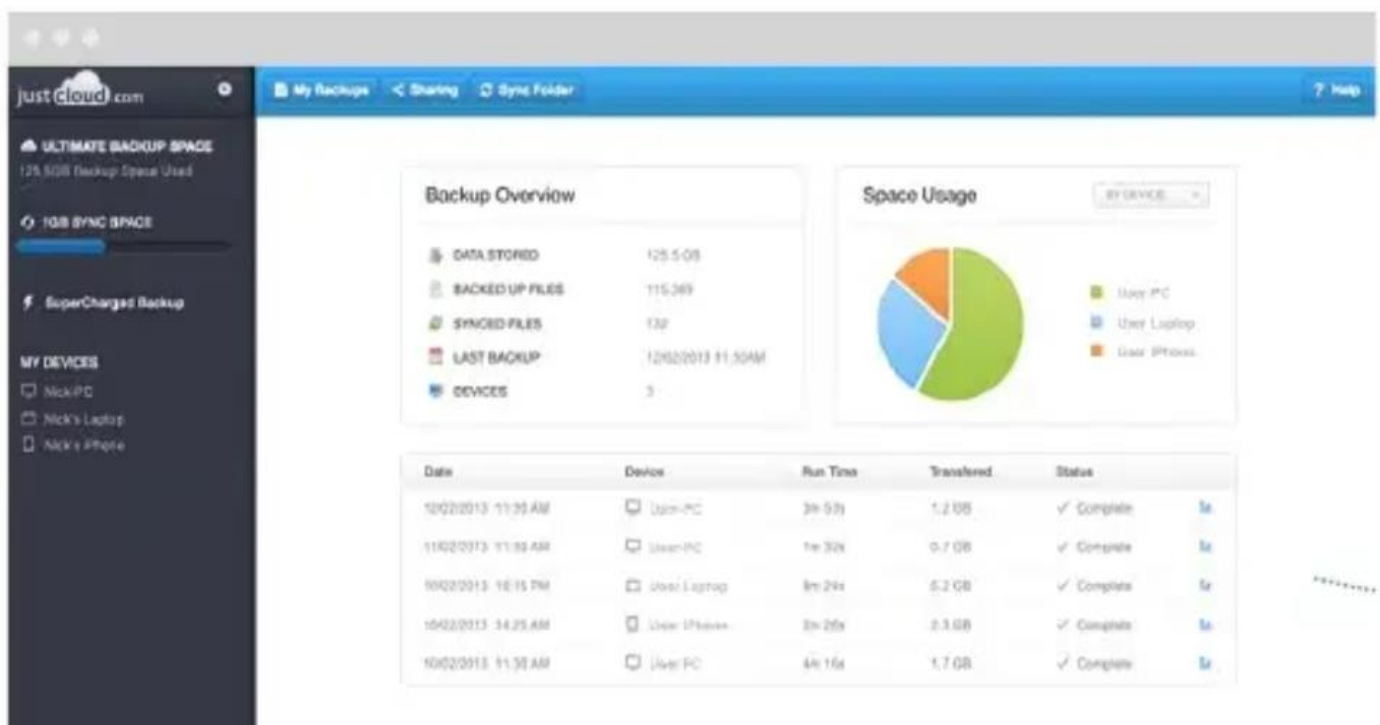
- ◆ JustCloud Trail:
- ◆ Visit the JustCloud website.
- ◆ Sign up for a free trial account.
- Installallation of JustCloud Application.

Features:

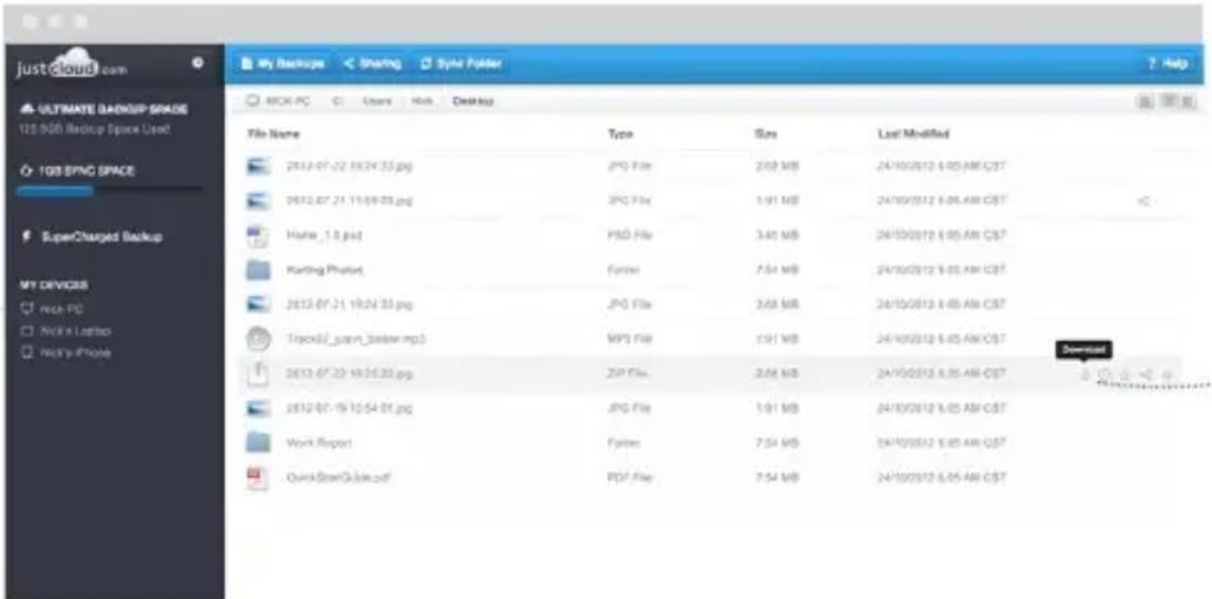
- ◆ File Synchronization : upload files to Justcloud and see how they Sync across devices.
- ◆ Back up Options: Explore the backup settings.
- ◆ Storage Limits: we should check the storage limits and performance during the trial period.

User Interface and Experience :

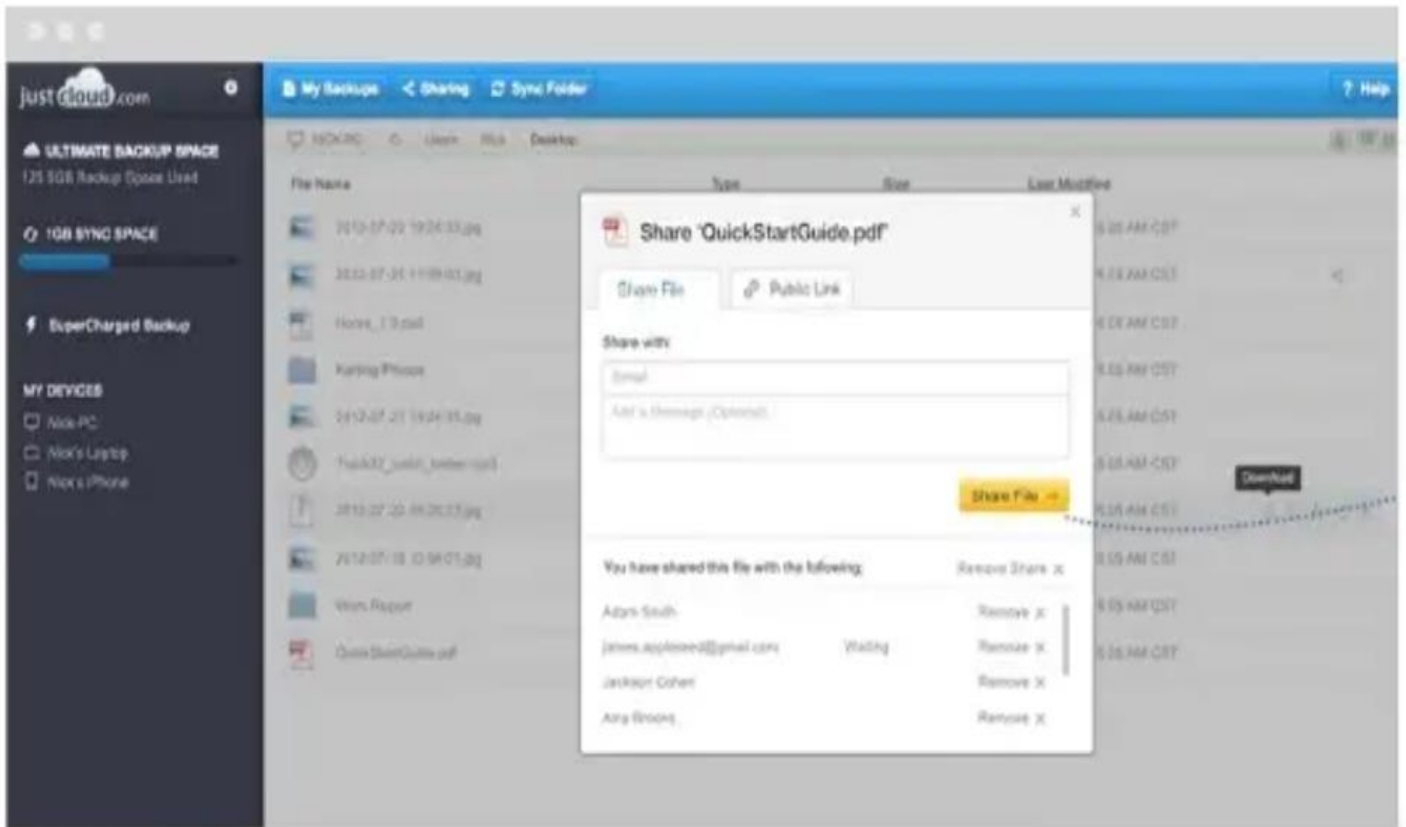
- ◆ Ultimate backup space.
- ◆ Automated backup.



- ◆ Backup app your device.
- ◆ We should access files anywhere.

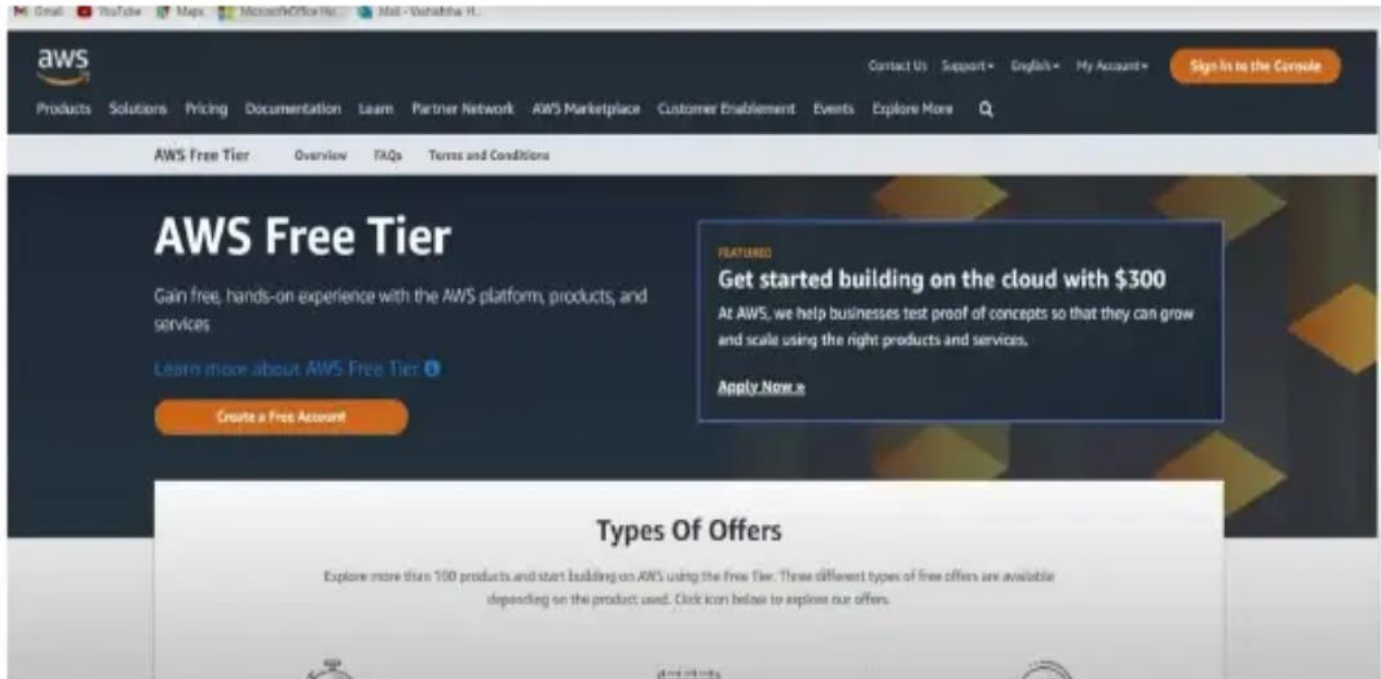


◆ We should share with friends.



AWS Solutions:

- ◆ Visit AWS portal and sign up.



Sign-up on AWS:

A screenshot of the 'Sign up for AWS' form. The AWS logo is at the top. The heading is 'Sign up for AWS'. The form includes a section for 'Explore Free Tier products with a new AWS account.' with a link 'To learn more, visit aws.amazon.com/free.' and an illustration of a hand holding three cubes. The main form fields are: 'Root user email address' (with a note 'Used for account recovery and some administrative functions'), 'AWS account name' (with a note 'Choose a name for your account. You can change this name in your account settings after you sign up.'), and a 'Verify email address' button. Below these is an 'OR' separator and a 'Sign in to an existing AWS account' button. An error message 'An AWS account name is required.' is displayed below the account name field.

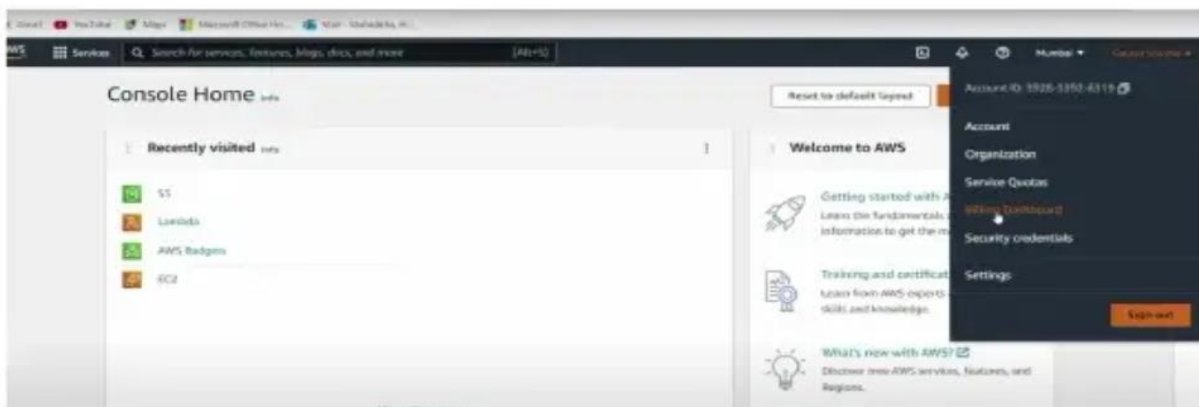
- ◆ Now we try free Version by adding Credit /debit card.

The screenshot shows the AWS Billing Information page. On the left, a 'Secure verification' box contains a message: 'We will not charge you for usage below AWS Free Tier limits. We may temporarily hold up to \$1 USD (or an equivalent amount in local currency) as a pending transaction for 3-5 days to verify your identity.' Below this is a shield icon with a checkmark. On the right, the 'Billing Information' section includes a 'Credit or Debit card number' field with a masked card image, an 'Expiration date' dropdown set to 'January 2028', and a 'Billing address' section with a radio button selected for 'Use my contact address'. At the bottom, there is a section titled 'Do you have a PAN?' with a brief explanation of the Permanent Account Number (PAN) and a field for the PAN number.

- ◆ Now we will explore free tier.

The screenshot shows the AWS Billing console. The left sidebar contains navigation links: Home, Billing, Orders and Invoices, Credits, Purchase orders, Cost & Usage Reports, Cost Categories, Cost allocation tags, Free Tier, Billing Conductor, Cost Management, Cost Explorer, Budgets, Budget Reports, Savings Plans, Preferences, and Billing preferences. The main content area is titled 'AWS Free Tier' and includes a 'Summary' section with a search bar. Below the search bar is a table with columns: Service, AWS Free Tier usage limit, Current usage, Forecasted usage, MTD actual usage %, and MTD. The table is currently empty, displaying the message 'There is no Free Tier services data available'.

- ◆ Now we will try billing dashboard in free tier.



Q3:
Use Google App Engine to write a program to multiply two matrices. Deploy it on the Google cloud.

Step1: Login:

Try Google Cloud Platform for free

Step 1 of 2

Alexis MP
notaleisemail@gmail.com [SWITCH ACCOUNT](#)

Country
France

Terms of Service
☒ I have read and agree to the [Google Cloud Platform Free Trial Terms of Service](#).
Required to continue

[CONTINUE](#)

Access to all Cloud Platform Products
Get everything you need to build and run your apps, websites and services, including Firebase and the Google Maps API.

\$300 credit for free
Put Google Cloud to work with \$300 in credit to spend over the next 90 days.

No autocharge after free trial ends
We ask you for your credit card to make sure you are not a robot. You won't be charged unless you manually upgrade to a paid account.

Step2: Use free trial and add payment method.

Secure verification

We will not charge you for usage below AWS Free Tier limits. We may temporarily hold up to \$1 USD (or an equivalent amount in local currency) as a pending transaction for 3-5 days to verify your identity.

Billing Information

Credit or Debit card number

Expiration date
January 2028

Billing address
☒ Use my contact address

Do you have a PAN?
Permanent Account Number (PAN) is a ten-digit alphanumeric number issued by the Indian Income Tax Department. This 10-digit number is printed on the front of your PAN card.

Step3: Start new project.

New Project

Project Name *

My Project 93540

Project ID *

animated-spider-220413

Project ID can have lowercase letters, digits, or hyphens. It must start with a lowercase letter and end with a letter or number.

Location *

 No organization

BROWSE

Parent organization or folder

CREATE

CANCEL

Write a program to multiply two matrices.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
  initial-scale=1.0">
  <title> Document </title>
</head>
<body>
```



```

<?php
$matrix1 = [
    [1, 2, 3]
    [4, 5, 6]
];
$matrix2 = [
    [7 , 8]
    [9 , 10],
    [ 11 , 12]
];
// now we will make a function to multiply tow matrix
function multiplymatrices ($matrix1 , $matrix2)
{
// now we will identify their rows and columns and their index value
$row1 = count ($matrix1);
$col1 = count ($matrix1 [0]);
$row2 = count ($matrix2);
$col2 = count ($matrix2[0]);
// Now checking of column in first column
matrix first row
if ($col1 != $row2) {
    echo "matrix Multiplication is not possible . the number of columns of matrix 1 must
        be equal to the number of rows
        .\n";
    return null;
}
// initialize the result matrix with zeros

```

```

        $result = array_fill (0 , $row1, array_fill(0, $col2, 0));

// initialize i , j , & k
        for ($i = 0; $i < $row1; $i++) {
            for ($j=0; $j< $col2; $j++) {
                // multiply elements with their index value
                $result[$i] [$j] += $matrix1[$i] [$k] *
                $matrix2 [$k] [$j];
            }
        }
        return $result;
    }

// Multiply the matrices (function call)

    $result = multiplymatrices($matrix1, $matrix2);

// display the result
if ($result !== null) {
    echo " Resultant Matrix after Multiplication: <br>";
    } }

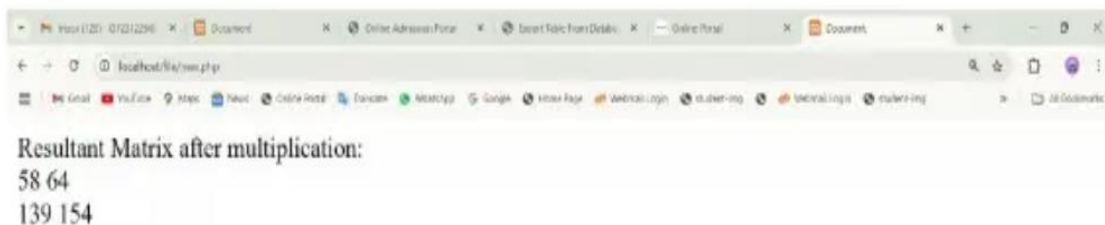
?>

</body>

</html>

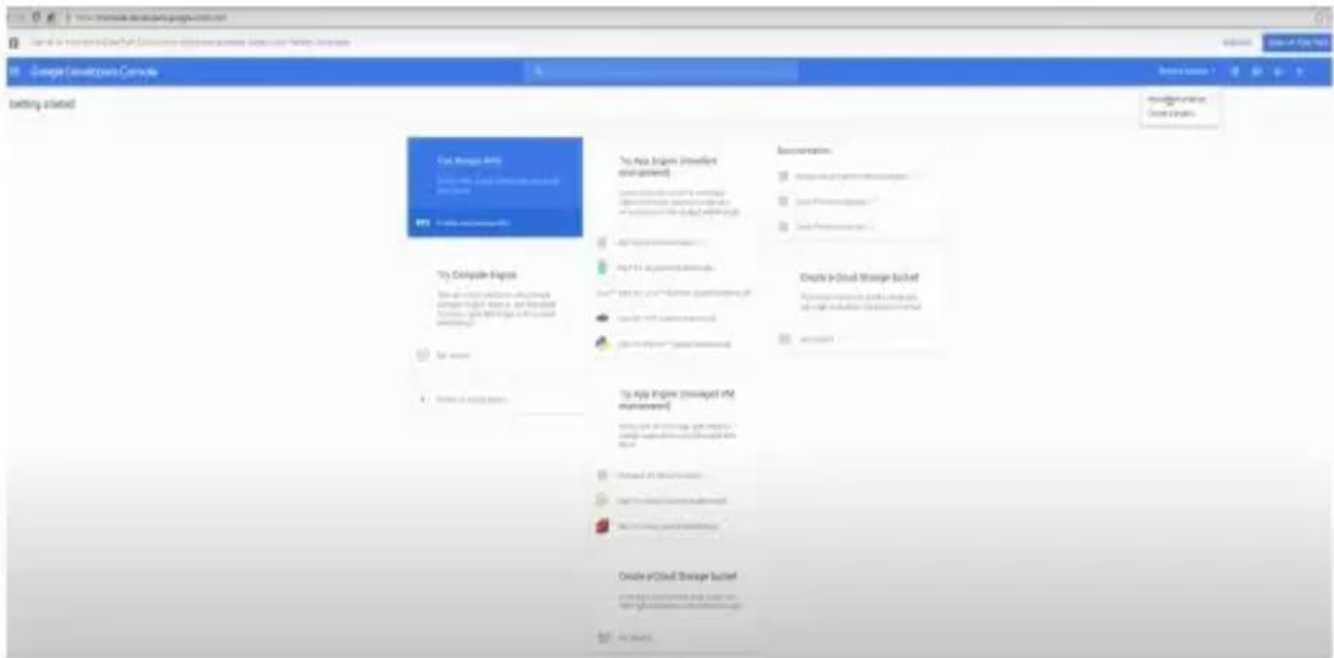
```

Output:

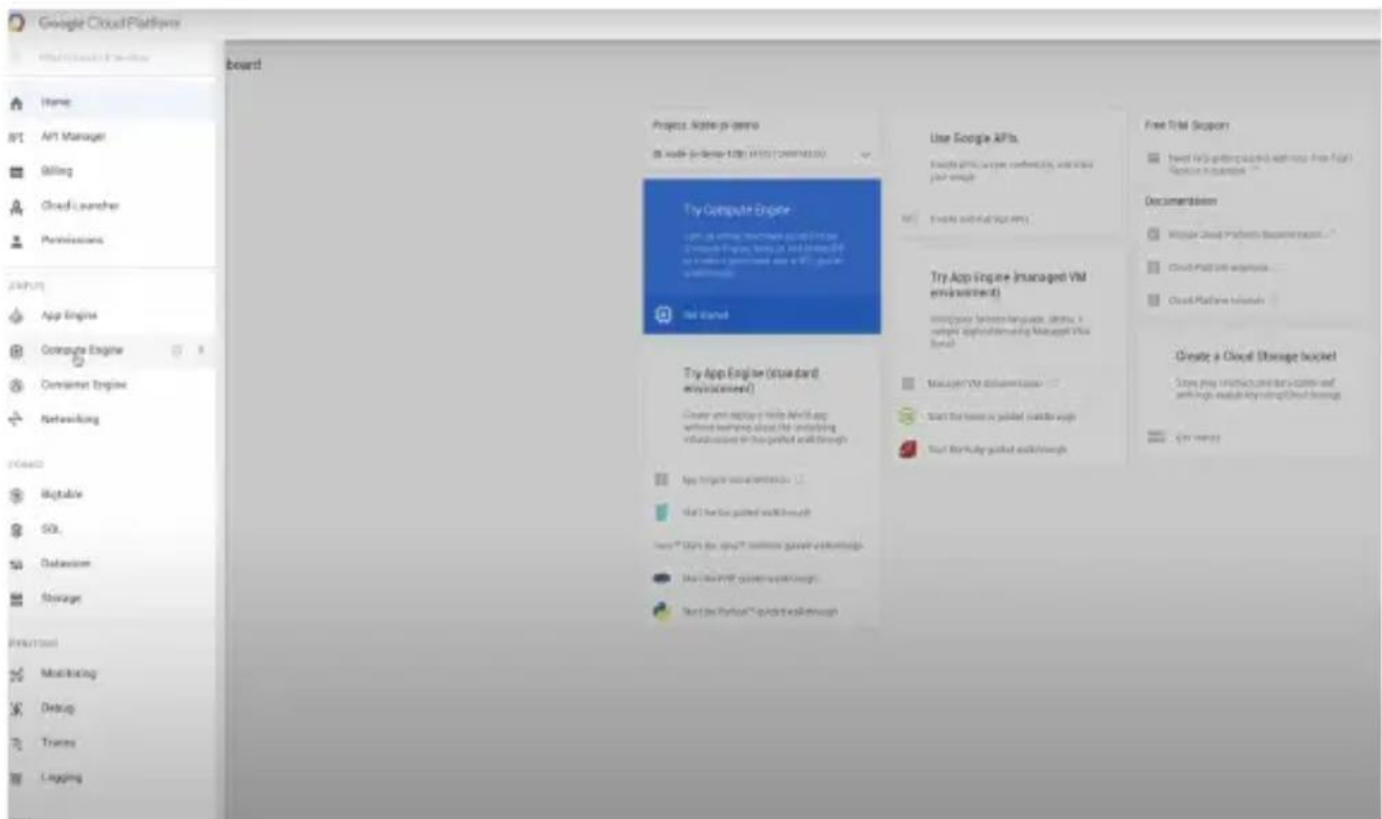


Deploy it on the google cloud.

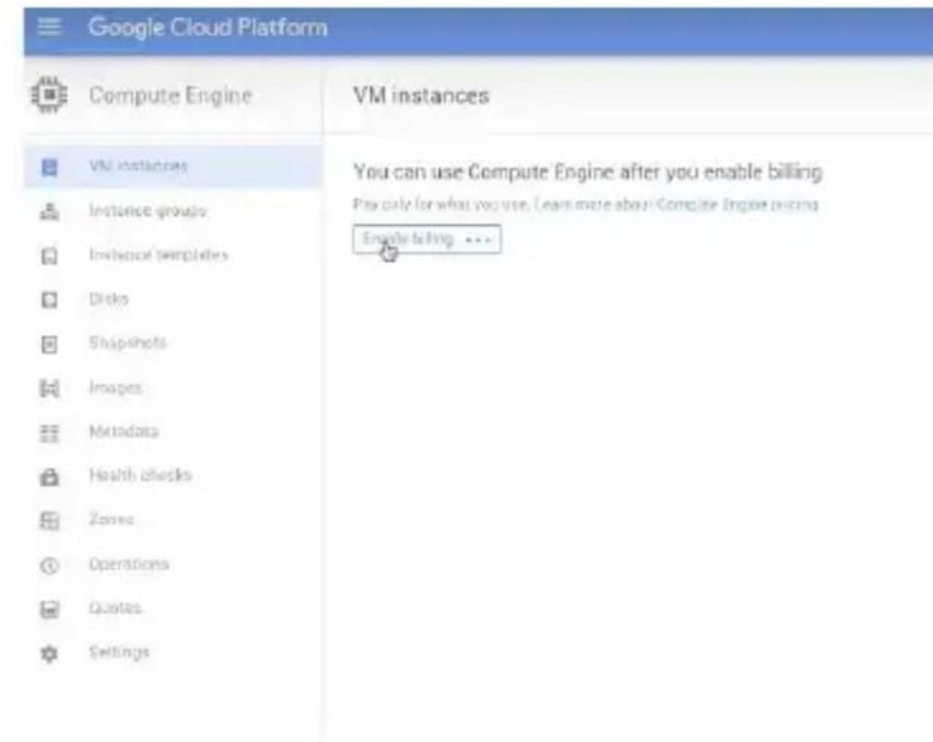
1. Click on Manage all project.



2. Click on Compute Engine.



3. Enable billing.



Section II: Data Science Lab:

Q1.

The height of 50 students of class X was measured in centimeters. The following table shows this data.

Perform the tasks (i) to (iv) using R programming.

#Height data

```
heights <- c(156, 155, 163, 151, 141, 160, 151, 131, 179, 163,  
            165, 160, 158, 168, 158, 154, 162, 166, 171, 141,  
            161, 159, 158, 156, 149, 145, 168, 171, 165, 145,  
            166, 167, 169, 156, 164, 146, 155, 164, 168, 172,  
            156, 163, 167, 157, 155, 145, 171, 164, 166, 195,)
```

(i) Minimum and maximum height

```
min_height <- min(heights)
min_height <- max(heights)
cat("minimum height:", min_height, "\n")
cat("maximum height:", max_height, "\n")
```

#(ii) Find the percentage of students whose Height is between 160 and 170 cm

```
count_160_170 <- sum(heights >= 160 & heights <= 170) * 100
cat("percentage of students with Height btw 160 and 170 cm:",
    percentage_160_170, "\n")
```

#(iii) create and draw the frequency distribution with a histogram

```
hist(heights, breaks=5, col="lightblue", main="height distribution",
     xlab="Height (cm)", ylab="frequency")
```

#(iv) Find the outlier of the data

```
boxplot(heights, main="boxplot of Heights", col="lightgreen",
     ylab="Height(cm)")
outliers <- boxplot.stats(heights)$out
cat("outliers in the data:", outliers, "\n")
```

Q2:

An organization collected the following data of its employees: “years in employment” and “employee’s salary”. Use R programming to fit a linear regression line to predict the effect of

years in employment on the employee's salary. Also , predict the salary of an employee who is employed for 11 years.

<u>Employee num</u>	<u>Years in Emp</u>	<u>Emp Monthly Sal in Thous Rs</u>
<u>1</u>	<u>10</u>	<u>149</u>
<u>2</u>	<u>8</u>	<u>120</u>
<u>3</u>	<u>12</u>	<u>150</u>
<u>4</u>	<u>7</u>	<u>110</u>
<u>5</u>	<u>6</u>	<u>100</u>
<u>6</u>	<u>5</u>	<u>90</u>
<u>7</u>	<u>9</u>	<u>130</u>
<u>8</u>	<u>8</u>	<u>110</u>
<u>9</u>	<u>3</u>	<u>80</u>
<u>10</u>	<u>4</u>	<u>85</u>

Now we will write the R program to fit a linear regression model and predict the salary for an employee with 11 years of employment:

```
#data
```

```
years <-c(10,8,12,7,6,5,9,8,3,4)
```

```
salary <-c(149,120,150,110,100,90,130,110,80,85)
```

```
#create a linear regression model
```

```
model <- lm(salary ~ years)
```

```
#Summary of the model
```

```
Summary(model)
```

```
#predict salary for 11 years of employment
```

```
predicted_salary <- predict(model, data.frame(years=11))
```

```
cat("predicted salary for 11 years of employment:", predicted_salary, "thousand Rs\n")
```

```
#Plot the data and regression line
```

```
plot(years, salary, main="years vs salary regression", xlab="Years in Employment".
```

```
ylab="Salary (in thousand Rs)", pch=16, col="blue")
```

```
abline(model, col="red")
```